



VICTORIA JUNIOR COLLEGE

JC 2 PRELIMINARY EXAMINATION 2018

NAME : _____

CT CLASS : _____

H2 BIOLOGY

9744/04

Paper 4

2 hours 30 minutes

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1. Write your name and CT group in the spaces at the top of this page.
 2. Write in dark blue or black pen. You may use an HB pencil for any diagrams or graphs.
 3. Answer all questions in the spaces provided on the Question Paper.
 4. Students with the microscope and slide **must start with Question 2 first**.
 5. The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.
 6. At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [] at the end of each question.

For Examiner's Use	
1	
2	
3	
Total	

This document consists of **14** printed pages, including the cover page.

Answer **all** questions.

1. You are required to investigate the effect of temperature on the activity of the enzyme urease. Urease catalyses the following reaction.



You should ensure that the reagents are mixed thoroughly by stirring with a glass rod before use.

Proceed as follows.

- 1 Prepare two beakers to act as water baths. The temperature of the water in one should be about 50 °C. In the other, the water should boil.
- 2 Label three test-tubes **A**, **B** and **C** respectively. Add 4 cm³ of urease into each test-tube. Place tube **A** in the water bath at 50 °C, tube **B** in the boiling water and tube **C** in a test-tube rack at room temperature.
- 3 After **five minutes**, remove tubes **A** and **B** from the water baths and cool them under a running tap.
- 4 Measure out 10 cm³ of urea solution into each of three clean test-tubes.
- 5 Adjust the temperature of one water bath to 38-40 °C, the other is no longer needed. Place tubes **A**, **B** and **C** and the three containing urea solution into this water bath.
- 6 After **three minutes**, tip the contents of one of the urea tubes into each of tubes **A**, **B** and **C**. Stir the contents using a glass rod. Note the time and leave these tubes in the water bath for **30 minutes**.

You should go on to another part of the question during the 30-minute period.

- 7 After **30 minutes**, remove tubes **A**, **B** and **C** from the water bath. Pour the contents of each tube into separate small beakers. Add five drops of indicator into each of the small beakers.
- (a) (i) With reference to the indicator colour chart (on screen), record the colour of the contents of the three test tubes, as well as the pH of the contents, in a suitable table.

[3]

(ii) Using your biological knowledge, explain your results.

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(iii) Explain the ways in which you would have expected the result to be different if tubes **A** and **B** had been maintained at their respective temperatures for one hour instead of five minutes.

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..... [2]

(iv) State a variable that is not controlled in your procedure that is likely to affect the rate of enzyme reaction.

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(v) Other than controlling the variable you mentioned in **(iv)**, using the same apparatus and materials, state **three** other ways in which you could improve the investigation into the effect of temperature on the activity of the enzyme urease.

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..... [3]

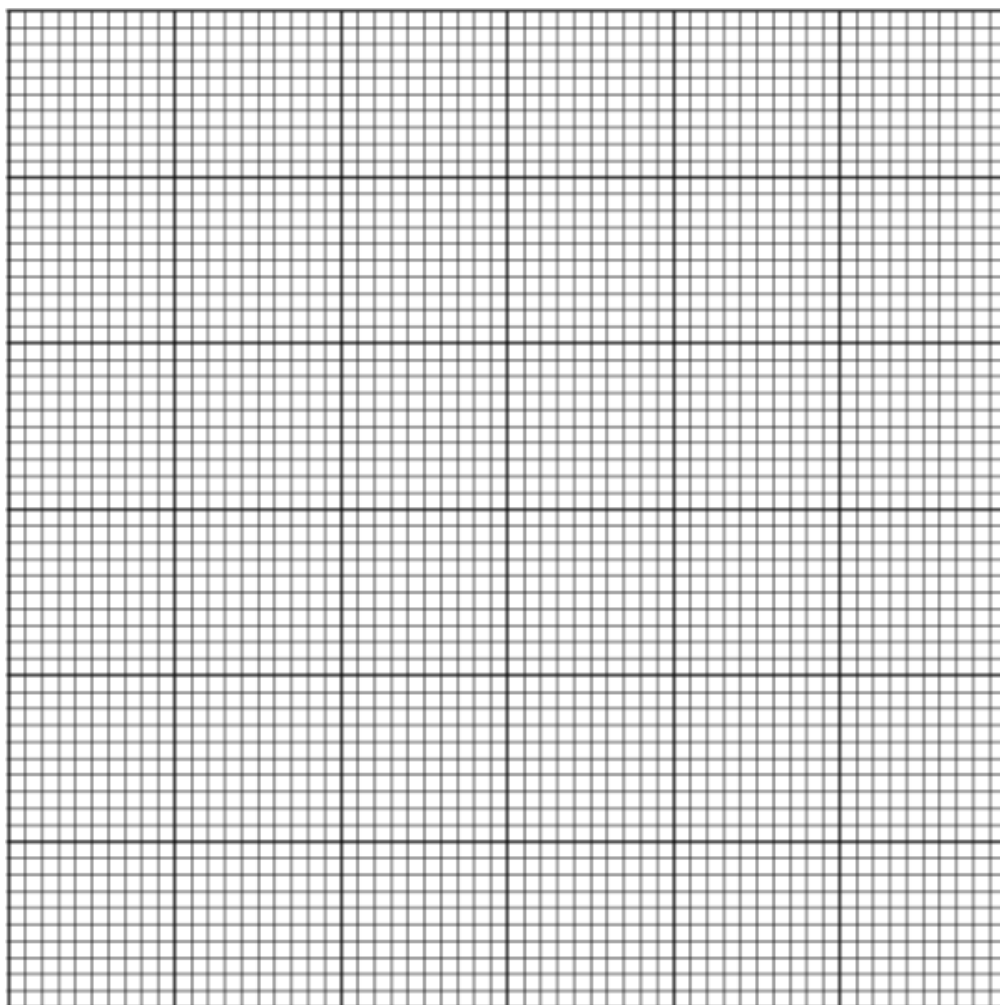
- (b)** Another student carried out a separate experiment to investigate the effect of urea concentration on the rate of enzymatic reaction. To find out the amount of ammonium carbonate produced in different urea concentrations, the student carried out titrations with dilute sulfuric acid and recorded down the volumes of acid needed to neutralise the ammonium carbonate.

The results are recorded in Table 1.1 below.

Urea concentration / mM	Volume of dilute sulfuric acid needed to neutralise the ammonium carbonate produced / cm ³			
	First run	Second run	Third run	Average
0.4	45	42	43	43.3
0.8	49	48	48	48.3
1.2	52	52	51	51.7
1.4	53	52	53	52.7
2.0	54	52	54	53.3

Table 1.1

- (i)** Plot a graph to show the effect of urea concentration on the rate of enzymatic reaction.



[4]

(ii) Comment on the effectiveness of the use of titrations with dilute sulfuric acid in the investigation as compared to the method used in **(a)**.

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..... [1]

The student then carried out a t-test to find out if there is a significant difference in the volume of dilute sulfuric acid needed when 0.4 nM of urea was used, compared to when 2.0 nM of urea was used. The p value was less than 0.05.

(iii) State the degree of freedom for the t-test.

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Explain what is meant by:

(iv) significant difference.

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(v) p value less than 0.05.

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[Total: 21]

2. During this question you will require access to a microscope.

You are required to investigate the effects of sodium nitrate and lead nitrate solutions on cells of the plant material with which you have been supplied.

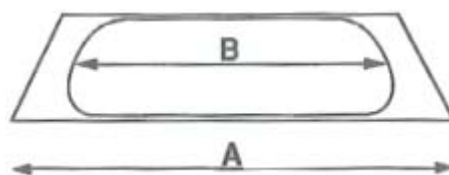
Proceed as follows.

- 1 Peel off one or two strips of epidermis from the most deeply pigmented areas of the plant tissue. Remove as little of the underlying tissue as possible. Cut the epidermis so that you have **three** squares of tissue, each about **5 mm x 5 mm**. Place these in a dish of distilled water.
- 2 Mount one piece of tissue on a microscope slide in **distilled water** under a cover slip.
- 3 Mount the other pieces in **1 mol dm⁻³ sodium nitrate solution** and **1 mol dm⁻³ lead nitrate solution**.
- 4 Label your slides appropriately.
- 5 Examine the tissue mounted in distilled water, using your microscope. Find an area of the tissue where pigmented cells occur, preferably as a single layer.

(a) Describe the distribution of the coloured contents within the cells.

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..... [1]

- 6 Observe the piece of tissue mounted in **1 mol dm⁻³ sodium nitrate solution**, using your microscope.
- 7 Decide which objective is the most appropriate for making measurements on the most deeply coloured cells which you can see. Use this objective to measure the dimensions of **A** and **B** of **five** of the cells immersed in 1 mol dm⁻³ sodium nitrate solution, as shown in Fig. 2.1.



A is the maximum length of the cell.

B is the maximum length of the cytoplasm.

Fig. 2.1

(b) (i) Record your measurements in 'graticule units' in an appropriate manner.

Objective lens used:

[4]

(ii) Account **fully** for the change in appearance of the cells when placed in 1 mol dm⁻³ sodium nitrate solution.

[4]

8 Observe the piece of tissue mounted in **1 mol dm⁻³ lead nitrate solution**, using your microscope.

(c) Describe the appearance of these cells compared with those in **step 6**.

..... [1]

- 9 Remove the cover slip from the slide with sodium nitrate solution. Blot up as much of the solution as possible. Add distilled water to the slide and replace the cover slip. **Immediately** and for several minutes, observe the appearance of pigmented cells. Repeat the procedure above on the cells mounted in lead nitrate solution.

- (d) (i) Record your observations of the cells that were originally mounted in sodium nitrate solution.

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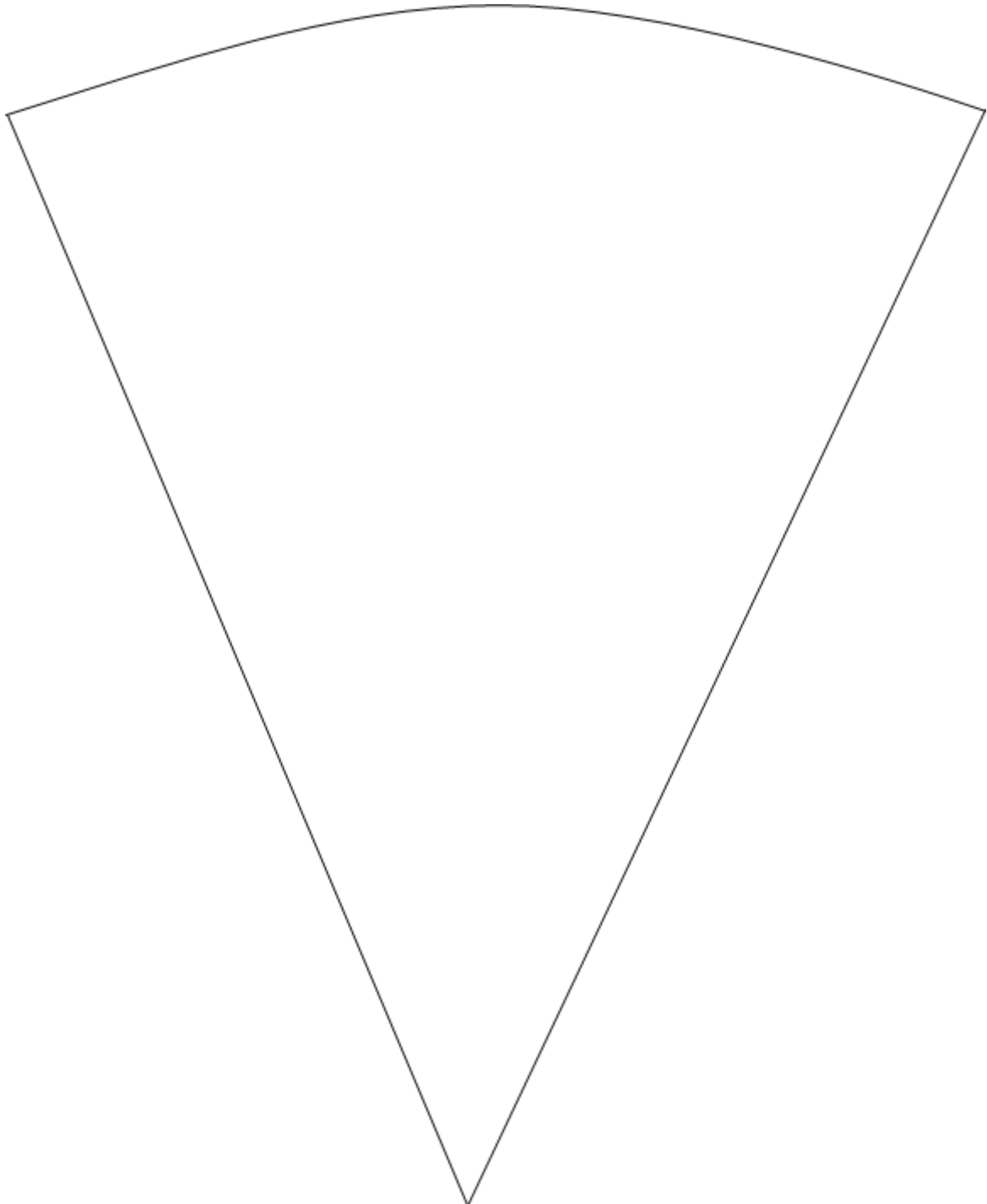
- (ii) Record your observations of the cells that were originally mounted in lead nitrate solution, pointing out any differences that occurred between these cells and those that were originally mounted in sodium nitrate solution.

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- (e) Explain your observations in (d)(i) and (ii) as fully as possible.

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- 10** Using a scalpel, cut a thin slice of the cross-section (as thinly as possible) of the plant material. Mount this in a drop of water on a microscope slide. Add a coverslip and examine the slide using your microscope.
- (f)** Draw a large, plan diagram of a segment of the plant material in the space provided.



[4]

[Total: 23]

3. Many people are intolerant to the disaccharide lactose, which is found in milk. The enzyme lactase is used commercially to catalyse the breakdown of lactose to the monosaccharides glucose and galactose. These sugars taste sweeter and are easier to digest than lactose.

Enzymes can be immobilised in a number of different ways, using different materials. Fig. 3.1 shows three ways of immobilisation of enzymes.

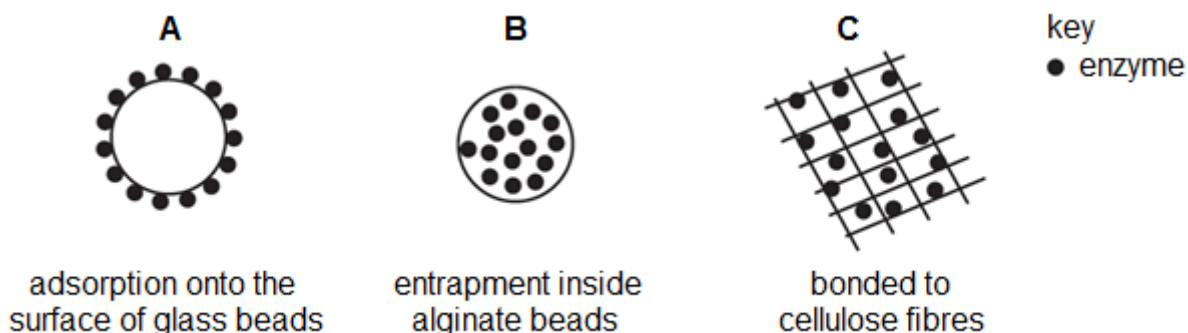


Fig. 3.1

A student carried out an investigation to compare the activity of the enzyme lactase that had been immobilised in the three different ways shown in Fig. 3.1.

- The immobilised enzymes were packed within a column.
- A solution containing 20 mg cm^{-3} of lactose was poured through the column containing the immobilised enzyme.
- The solution containing the products was collected and the concentration of product was measured.
- Presence of galactose can be detected through oxidation by galactose oxidase which results in a colorimetric 570nm product (a colorimetric product that can be detected at a wavelength of 570nm), proportional to the galactose present.

Outline a procedure the student could use to compare the activity of lactase that has been immobilised in different ways.

In your plan, you must use:

- a solution of 20 mg cm^{-3} of lactose
- glass beads with 1 M lactase adsorbed on it
- alginate-entrapped 1 M lactase
- cellulose-bound 1 M lactase
- glass beads
- alginate beads
- cellulose
- galactose oxidase
- burettes
- retort stands

You may select from the following apparatus in the design of your experiment:

- normal laboratory glassware e.g. glass rods, test tubes, boiling tubes, beakers, measuring cylinders, and pipettes
- syringes
- timer e.g. stopwatch
- mesh
- thermostatically controlled water baths
- 1 M hydrochloric acid
- Benedict's reagent
- Biuret solution
- test tube holders
- colorimeter and cuvettes

Your plan should:

- have a clear and helpful structure such that the method you used is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary
- identify the dependent and independent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- propose a statistical test that can be used to determine if the results obtained are significant.
- use the correct technical and scientific terms
- include reference to safety measures to minimize any risks associated with the proposed experiment.

[Total: 11]

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